

Wei D. Lu

professor

基本信息

姓名: Wei D. Lu

所属院校: 密歇根大学-安娜堡

地区: United States

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

人工智能

电子与计算机工程

材料物理

集成电路设计与集成系统

研究领域

深度学习与神经网络

嵌入式系统与硬件设计

存储器设计与架构

计算机体系结构

薄膜材料与表面物理

纳米物理与纳米器件

纳米电子与先进工艺

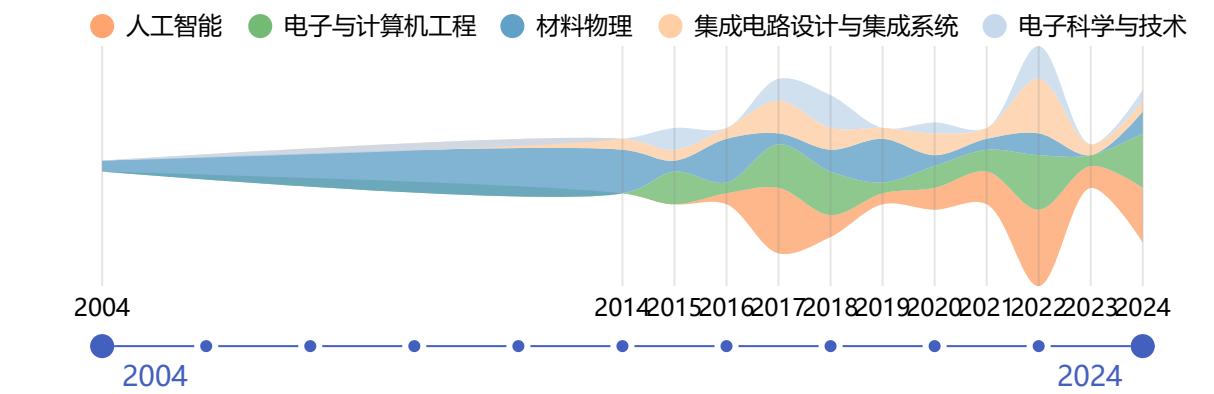
个人简介

Wei D. Lu, also known as Wei Lu, is the James R. Mellor Professor in the Department of Electrical Engineering and Computer Science at the University of Michigan. He is an IEEE Fellow and co-founder of MemryX Inc. and Crossbar Inc. His research focuses on nanoelectronics, memory, memristor, neuromorphic AI chips, and related fields. With an h-index of 92, he has significantly contributed to the advancement of memristive systems and neuromorphic computing.

工作经历

Univ of Michigan - James R. Mellor Professor	-
MemryX Inc - co - founder	-
Crossbar Inc - co - founder	-

近年发表的论文



TT-CIM: Tensor Train Decomposition for Neural Network in RRAM-Based Compute-in-Memory Systems
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS
Meng, Fan-Hsuan; Wu, Yuting; Zhang, Zhengya; Lu, Wei D.
被引用次数: 1 2024

Thermodynamic origin of nonvolatility in resistive memory
MATTER
Li, Jingxian; Appachar, Anirudh; Peczonczyk, Sabrina L.; Harrison, Elisa T.; Ievlev, Anton V.; Hood, Ryan; Shin, Dongjae; Yoo, Sangmin; Roest, Brianna; Sun, Kai; Beckmann, Karsten; Popova, Olya; Chiang, Tony; Wahby, William S.; Jacobs-Godrim, Robin B.; Marinella, Matthew J.; Maksymovych, Petro; Heron, John T.; Cady, Nathaniel; Lu, Wei D.; Kumar, Suhas; Talin, A. Alec; Sun, Wenhao; Li, Yiyang
被引用次数: 0 2024

Side-Channel Attack Analysis on In-Memory Computing Architectures
IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING
Wang, Ziyu; Meng, Fan-Hsuan; Park, Yongmo; Eshraghian, Jason K.; Lu, Wei D.
被引用次数: 5 2024

Exploiting deep learning accelerators for neuromorphic workloads
NEUROMORPHIC COMPUTING AND ENGINEERING
Sun, Pao-Sheng Vincent; Titterton, Alexander; Gopiani, Anjlee; Santos, Tim; Basu, Arindam; Lu, Wei D.; Eshraghian, Jason K.
被引用次数: 2 2024

Safe, secure and trustworthy compute-in-memory accelerators

NATURE ELECTRONICS

Wang, Ziyu; Wu, Yuting; Park, Yongmo; Lu, Wei D.

被引用次数: 0

2024

Compute-In-Memory Technologies for Deep Learning Acceleration

IEEE NANOTECHNOLOGY MAGAZINE

Meng, Fan-husan; Lu, Wei D.

被引用次数: 5

2024

Editorial: Focus issue on in-memory computing

NEUROMORPHIC COMPUTING AND ENGINEERING

Lu, Wei D.; Payvand, Melika; Yang, Yuchao

被引用次数: 0

2024

RN-Net: Reservoir Nodes-Enabled Neuromorphic Vision Sensing Network

ADVANCED INTELLIGENT SYSTEMS

Yoo, Sangmin; Lee, Eric Yeu-Jer; Wang, Ziyu; Wang, Xinxin; Lu, Wei D.

被引用次数: 1

2024

Interspecies-chimera machine vision with polarimetry for real-time navigation and anti-glare pattern recognition

NATURE COMMUNICATIONS

Guo, Tao; Li, Shasha; Zhou, Y. Norman; Lu, Wei D.; Yan, Yong; Wu, Yimin A.

被引用次数: 0

2024

Efficient data processing using tunable entropy-stabilized oxide memristors

NATURE ELECTRONICS

Yoo, Sangmin; Chae, Sieun; Chiang, Tony; Webb, Matthew; Ma, Tao; Paik, Hanjong; Park, Yongmo; Williams, Logan; Nomoto, Kazuki; Xing, Huili G.; Trolrier-McKinstry, Susan; Kioupakis, Emmanouil; Heron, John T.; Lu, Wei D.

被引用次数: 4

2024

Training Spiking Neural Networks Using Lessons From Deep Learning

PROCEEDINGS OF THE IEEE

Eshraghian, Jason K.; Ward, Max; Neftci, Emre O.; Wang, Xinxin; Lenz, Gregor; Dwivedi, Girish; Bennamoun, Mohammed; Jeong, Doo Seok; Lu, Wei D.

被引用次数: 109

2023

PowerGAN: A Machine Learning Approach for Power Side-Channel Attack on Compute-in-Memory Accelerators

ADVANCED INTELLIGENT SYSTEMS

Wang, Ziyu; Wu, Yuting; Park, Yongmo; Yoo, Sangmin; Wang, Xinxin; Eshraghian, Jason K.; Lu, Wei D.

被引用次数: 2

2023

Bulk-Switching Memristor-Based Compute-In-Memory Module for Deep Neural Network Training

ADVANCED MATERIALS

Wu, Yuting; Wang, Qiwen; Wang, Ziyu; Wang, Xinxin; Ayyagari, Buvna; Krishnan, Siddarth; Chudzik, Michael; Lu, Wei D.

被引用次数: 3

2023

TAICHI: A Tiled Architecture for In-Memory Computing and Heterogeneous Integration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Wang, Xinxin; Pinkham, Reid; Zidan, Mohammed A.; Meng, Fan-Hsuan; Flynn, Michael P.; Zhang, Zhengya; Lu, Wei D.

被引用次数: 9

2022

Physical Unclonable Function Systems Based on Pattern Transfer of Fingerprint-Like Patterns

IEEE ELECTRON DEVICE LETTERS

Wang, Ziyu; Zhu, Xiaojian; Jeloka, Supreet; Cline, Brian; Lu, Wei D.

被引用次数: 2

2022

Device Variation Effects on Neural Network Inference Accuracy in Analog In-Memory Computing Systems

ADVANCED INTELLIGENT SYSTEMS

Wang, Qiwen; Park, Yongmo; Lu, Wei D.

被引用次数: 16

2022

Columnar Learning Networks for Multisensory Spatiotemporal Learning

ADVANCED INTELLIGENT SYSTEMS

Yoo, Sangmin; Park, Yongmo; Wang, Ziyu; Wu, Yuting; Medepalli, Saaketh; Thio, Wesley; Lu, Wei D.

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2022

Tuning Resistive Switching Behavior by Controlling Internal Ionic Dynamics for Biorealistic Implementation of Synaptic Plasticity

ADVANCED ELECTRONIC MATERIALS

Yoo, Sangmin; Wu, Yuting; Park, Yongmo; Lu, Wei D.

被引用次数: 12

2022

Investigating Selectorless Property within Niobium Devices for Storage Applications

ACS APPLIED MATERIALS & INTERFACES

Chen, Po-Hsun; Lin, Chih-Yang; Chang, Ting-Chang; Eshraghian, Jason K.; Chao, Yu-Ting; Lu, Wei D.; Sze, Simon M.

被引用次数: 17

2022

Dynamical memristors for higher-complexity neuromorphic computing

NATURE REVIEWS MATERIALS

Kumar, Suhas; Wang, Xinxin; Strachan, John Paul; Yang, Yuchao; Lu, Wei D.

被引用次数: 198

2022

Gradient-Based Neuromorphic Learning on Dynamical RRAM Arrays

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Zhou, Peng; Choi, Dong-Uk; Lu, Wei D.; Kang, Sung-Mo; Eshraghian, Jason K.

被引用次数: 16

2022

Exploring Compute-in-Memory Architecture Granularity for Structured Pruning of Neural Networks

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Meng, Fan-Hsuan; Wang, Xinxin; Wang, Ziyu; Lee, Eric Yeu-Jer; Lu, Wei D.

被引用次数: 7

2022

Design Space Exploration of Dense and Sparse Mapping Schemes for RRAM Architectures

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Lammie, Corey; Eshraghian, Jason K.; Li, Chenqi; Amirsoleimani, Amirali; Genov, Roman; Lu, Wei D.; Azghadi, Mostafa Rahimi

被引用次数: 0

2022

Dynamic resistive switching devices for neuromorphic computing

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Wu, Yuting; Wang, Xinxin; Lu, Wei D.

被引用次数: 11

2022

Spatiotemporal Spike Pattern Detection with Second-order Memristive Synapses

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Wu, Yuting; Yoo, Sangmin; Meng, Fan-hsuan; Lu, Wei D.

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2022

Memristor-Based Binarized Spiking Neural Networks

IEEE NANOTECHNOLOGY MAGAZINE

Eshraghian, Jason K.; Wang, Xinxin; Lu, Wei D.

被引用次数: 52

2022

Memristive technologies for data storage, computation, encryption, and radio-frequency communication

SCIENCE

Lanza, Mario; Sebastian, Abu; Lu, Wei D.; Le Gallo, Manuel; Chang, Meng-Fan; Akinwande, Deji; Puglisi, Francesco M.; Alshareef, Husam N.; Liu, Ming; Roldan, Juan B.

被引用次数: 267

2022

How to Build a Memristive Integrate-and-Fire Model for Spiking Neuronal Signal Generation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kang, Sung Mo; Choi, Donguk; Eshraghian, Jason K.; Zhou, Peng; Kim, Jieun; Kong, Bai-Sun; Zhu, Xiaojian; Demirkol, Ahmet Samil; Ascoli, Alon; Tetzlaff, Ronald; Lu, Wei D.; Chua, Leon O.

被引用次数: 27

2021

RRAM-enabled AI Accelerator Architecture

2021 IEEE INTERNATIONAL ELECTRON DEVICES MEETING (IEDM)

Wang, Xinxin; Wu, Yuting; Lu, Wei D.

被引用次数: 0

2021

Memristors Based on (Zr, Hf, Nb, Ta, Mo, W) High-Entropy Oxides

ADVANCED ELECTRONIC MATERIALS

Ahn, Minhyung; Park, Yongmo; Lee, Seung Hwan; Chae, Sieun; Lee, Jihang; Heron, John T.; Kioupakis, Emmanouil; Lu, Wei D.; Phillips, Jamie D.

被引用次数: 29

2021

Neural connectivity inference with spike-timing dependent plasticity network

SCIENCE CHINA-INFORMATION SCIENCES

Moon, John; Wu, Yuting; Zhu, Xiaojian; Lu, Wei D.

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2021

Neural Functional Connectivity Reconstruction with Second-Order Memristor Network

ADVANCED INTELLIGENT SYSTEMS

Wu, Yuting; Moon, John; Zhu, Xiaojian; Lu, Wei D.

被引用次数: 10

2021

Memristive Stochastic Computing for Deep Learning Parameter Optimization

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lammie, Corey; Eshraghian, Jason K.; Lu, Wei D.; Azghadi, Mostafa Rahimi

被引用次数: 29

2021

A high-speed MIM resistive memory cell with an inherent vanadium selector

APPLIED MATERIALS TODAY

Lin, Chih-Yang; Tseng, Yi-Ting; Chen, Po-Hsun; Chang, Ting-Chang; Eshraghian, Jason K.; Wang, Qiwen; Lin, Qi; Tan, Yung-Fang; Tai, Mao-Chou; Hung, Wei-Chun; Huang, Hui-Chun; Lu, Wei D.; Sze, Simon M.

被引用次数: 12

2020

Memristor networks for real-time neural activity analysis

NATURE COMMUNICATIONS

Zhu, Xiaojian; Wang, Qiwen; Lu, Wei D.

被引用次数: 144

2020

Power-efficient combinatorial optimization using intrinsic noise in memristor Hopfield neural networks

NATURE ELECTRONICS

Cai, Fuxi; Kumar, Suhas; Van Vaerenbergh, Thomas; Sheng, Xia; Liu, Rui; Li, Can; Liu, Zhan; Foltin, Martin; Yu, Shimeng; Xia, Qiangfei; Yang, J. Joshua; Beausoleil, Raymond; Lu, Wei D.; Strachan, John Paul

被引用次数: 236

2020

A Crossbar-Based In-Memory Computing Architecture

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Xinxin; Zidan, Mohammed A.; Lu, Wei D.

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2020

Adaptive Synaptic Memory via Lithium Ion Modulation in RRAM Devices

SMALL

Lin, Chih-Yang; Chen, Jia; Chen, Po-Hsun; Chang, Ting-Chang; Wu, Yuting; Eshraghian, Jason K.; Moon, John; Yoo, Sangmin; Wang, Yu-Hsun; Chen, Wen-Chung; Wang, Zhi-Yang; Huang, Hui-Chun; Li, Yi; Miao, Xiangshui; Lu, Wei D.; Sze, Simon M.

被引用次数: 57

2020

Filament-Free Bulk Resistive Memory Enables Deterministic Analogue Switching

ADVANCED MATERIALS

Li, Yiyang; Fuller, Elliot J.; Sugar, Joshua D.; Yoo, Sangmin; Ashby, David S.; Bennett, Christopher H.; Horton, Robert D.; Bartsch, Michael S.; Marinella, Matthew J.; Lu, Wei D.; Talin, A. Alec

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2020

Nanoscale resistive switching devices for memory and computing applications

NANO RESEARCH

Lee, Seung Hwan; Zhu, Xiaojian; Lu, Wei D.

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2020

Stabilization of Mode-Dependent Impulsive Hybrid Systems Driven by DFA With Mixed-Mode Effects

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Zhang, Junhui; Li, Anni; Lu, Wei D.; Sun, Jitao

被引用次数: 14

2020

Near infrared neuromorphic computing via upconversion-mediated optogenetics

NANO ENERGY

Zhai, Yongbiao; Zhou, Ye; Yang, Xueqing; Wang, Feng; Ye, Wenbin; Zhu, Xiaojian; She, Donghong; Lu, Wei D.; Han, Su-Ting

被引用次数: 48

2020

Quantitative, Dynamic TaOx Memristor/Resistive Random Access Memory Model

ACS APPLIED ELECTRONIC MATERIALS

Lee, Seung Hwan; Moon, John; Jeong, YeonJoo; Lee, Jihang; Li, Xinyi; Wu, Huaqiang; Lu, Wei D.

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2020

Nanoionic Resistive-Switching Devices

ADVANCED ELECTRONIC MATERIALS

Zhu, Xiaojian; Lee, Seung Hwan; Lu, Wei D.

被引用次数: 49

2019

Recommended Methods to Study Resistive Switching Devices

ADVANCED ELECTRONIC MATERIALS

Lanza, Mario; Wong, H-S Philip; Pop, Eric; Ielmini, Daniele; Strukov, Dimitri; Regan, Brian C.; Larcher, Luca; Villena, Marco A.; Yang, J. Joshua; Goux, Ludovic; Belmonte, Attilio; Yang, Yuchao; Puglisi, Francesco M.;

Kang, Jinfeng; Magyari-Kope, Blanka; Yalon, Eilam; Kenyon, Anthony; Buckwell, Mark; Mehonic, Adnan; Shluger, Alexander; Li, Haitong; Hou, Tuo-Hung; Hudec, Boris; Akinwande, Deji; Ge, Ruijing; Ambrogio, Stefano; Roldan, Juan B.; Miranda, Enrique; Sune, Jordi; Pey, Kin Leong; Wu, Xing; Raghavan, Nagarajan; Wu, Ernest; Lu, Wei D.; Navarro, Gabriele; Zhang, Weidong; Wu, Huaqiang; Li, Runwei; Holleitner, Alexander; Wurstbauer, Ursula; Lemme, Max C.; Liu, Ming; Long, Shibing; Liu, Qi; Lv, Hangbing; Padovani, Andrea; Pavan, Paolo; Valov, Ilia; Jing, Xu; Han, Tingting; Zhu, Kaichen; Chen, Shaochuan; Hui, Fei; Shi, Yuanyuan

被引用次数: 463

2019

A fully integrated reprogrammable memristor-CMOS system for efficient multiply-accumulate operations

NATURE ELECTRONICS

Cai, Fuxi; Correll, Justin M.; Lee, Seung Hwan; Lim, Yong; Bothra, Vishishtha; Zhang, Zhengya; Flynn, Michael P.; Lu, Wei D.

被引用次数: 527

2019

Temporal data classification and forecasting using a memristor-based reservoir computing system

NATURE ELECTRONICS

Moon, John; Ma, Wen; Shin, Jong Hoon; Cai, Fuxi; Du, Chao; Lee, Seung Hwan; Lu, Wei D.

被引用次数: 384

2019

Charge Transition of Oxygen Vacancies during Resistive Switching in Oxide-Based RRAM

ACS APPLIED MATERIALS & INTERFACES

Lee, Jihang; Schell, William; Zhu, Xiaojian; Kioupakis, Emmanouil; Lu, Wei D.

被引用次数: 98

2019

In Situ Nano-thermomechanical Experiment Reveals Brittle to Ductile Transition in Silicon Nanowires

NANO LETTERS

Cheng, Guangming; Zhang, Yin; Chang, Tzu-Hsuan; Liu, Qunfeng; Chen, Lin; Lu, Wei D.; Zhu, Ting; Zhu, Yong

被引用次数: 26

2019

Ionic modulation and ionic coupling effects in MoS₂ devices for neuromorphic computing

NATURE MATERIALS

Zhu, Xiaojian; Li, Da; Liang, Xiaogan; Lu, Wei D.

被引用次数: 482

2019

A general memristor-based partial differential equation solver

NATURE ELECTRONICS

Zidan, Mohammed A.; Jeong, YeonJoo; Lee, Jihang; Chen, Bing; Huang, Shuo; Kushner, Mark J.; Lu, Wei D.

被引用次数: 200

2018

The future of electronics based on memristive systems

NATURE ELECTRONICS

Zidan, Mohammed A.; Strachan, John Paul; Lu, Wei D.

被引用次数: 1544

2018

MoS2 Memristors Exhibiting Variable Switching Characteristics toward Biorealistic Synaptic Emulation

ACS NANO

Li, Da; Wu, Bin; Zhu, Xiaojian; Wang, Juntong; Ryu, Byunghoon; Lu, Wei D.; Lu, Wei; Liang, Xiaogan

被引用次数: 203

2018

Optogenetics-Inspired Tunable Synaptic Functions in Memristors

ACS NANO

Zhu, Xiaojian; Lu, Wei D.

被引用次数: 229

2018

Self-Limited and Forming-Free CBRAM Device With Double Al₂O₃ ALD Layers

IEEE ELECTRON DEVICE LETTERS

Shin, Jong Hoon; Wang, Qiwen; Lu, Wei D.

被引用次数: 24

2018

On-Demand Reconfiguration of Nanomaterials: When Electronics Meets Ionics

ADVANCED MATERIALS

Lee, Jihang; Lu, Wei D.

被引用次数: 172

2018

Neuromorphic computing with memristive devices

SCIENCE CHINA-INFORMATION SCIENCES

Ma, Wen; Zidan, Mohammed A.; Lu, Wei D.

被引用次数: 43

2018

Parasitic Effect Analysis in Memristor-Array-Based Neuromorphic Systems

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Jeong, YeonJoo; Zidan, Mohammed A.; Lu, Wei D.

被引用次数: 78

2018

K-means Data Clustering with Memristor Networks

NANO LETTERS

Jeong, YeonJoo; Lee, Jihang; Moon, John; Shin, Jong Hoon; Lu, Wei D.

被引用次数: 97

2018

Ge nanowire photodetector with high photoconductive gain epitaxially integrated on Si substrate

APPLIED PHYSICS LETTERS

Otuonye, Ugo; Kim, Hee Woo; Lu, Wei D.

被引用次数: 45

2017

Temporal Learning Using Second-Order Memristors

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Zidan, Mohammed A.; Jeong, YeonJoo; Lu, Wei D.

被引用次数: 28

2017

Reservoir computing using dynamic memristors for temporal information processing

NATURE COMMUNICATIONS

Du, Chao; Cai, Fuxi; Zidan, Mohammed A.; Ma, Wen; Lee, Seung Hwan; Lu, Wei D.

被引用次数: 686

2017

Multifunctional Nanoionic Devices Enabling Simultaneous Heterosynaptic Plasticity and Efficient In-Memory Boolean Logic

ADVANCED ELECTRONIC MATERIALS

Yang, Yuchao; Yin, Minghui; Yu, Zhizhen; Wang, Zongwei; Zhang, Teng; Cai, Yimao; Lu, Wei D.; Huang, Ru

被引用次数: 57

2017

Sparse coding with memristor networks

NATURE NANOTECHNOLOGY

Sheridan, Patrick M.; Cai, Fuxi; Du, Chao; Ma, Wen; Zhang, Zhengya; Lu, Wei D.

被引用次数: 520

2017

Experimental Demonstration of Feature Extraction and Dimensionality Reduction Using Memristor Networks

NANO LETTERS

Choi, Shinhyun; Shin, Jong Hoon; Lee, Jihang; Sheridan, Patrick; Lu, Wei D.

被引用次数: 167

2017

Electronic and optical properties of oxygen vacancies in amorphous Ta₂O₅ from first principles

NANOSCALE

Lee, Jihang; Lu, Wei D.; Kioupakis, Emmanouil

被引用次数: 46

2017

Emulation of synaptic metaplasticity in memristors

NANOSCALE

Zhu, Xiaojian; Du, Chao; Jeong, YeonJoo; Lu, Wei D.

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2017

Iodine Vacancy Redistribution in Organic-Inorganic Halide Perovskite Films and Resistive Switching Effects

ADVANCED MATERIALS

Zhu, Xiaojian; Lee, Jihang; Lu, Wei D.

被引用次数: 296

2017

Memristive computing devices and applications

JOURNAL OF ELECTROCERAMICS

Zidan, Mohammed A.; Chen, An; Indiveri, Giacomo; Lu, Wei D.

被引用次数: 44

2017

Vertical Ge/Si Core/Shell Nanowire Junctionless Transistor

NANO LETTERS

Chen, Lin; Cai, Fuxi; Otuonye, Ugo; Lu, Wei D.

被引用次数: 128

2016

In Situ Nanoscale Electric Field Control of Magnetism by Nanoionics

ADVANCED MATERIALS

Zhu, Xiaojian; Zhou, Jiantao; Chen, Lin; Guo, Shanshan; Liu, Gang; Li, Run-Wei; Lu, Wei D.

被引用次数: 59

2016

Progress in the Characterizations and Understanding of Conducting Filaments in Resistive Switching Devices

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Yang, Yuchao; Lu, Wei D.

被引用次数: 29

2016

Electrochemistry at the Nanoscale

NANOSCALE

Valov, Ilia; Lu, Wei D.

被引用次数: 3

2016

Nanoscale electrochemistry using dielectric thin films as solid electrolytes

NANOSCALE

Valov, Ilia; Lu, Wei D.

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2016

Feature Extraction Using Memristor Networks

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Sheridan, Patrick M.; Du, Chao; Lu, Wei D.

被引用次数: 68

2016

Tuning Ionic Transport in Memristive Devices by Graphene with Engineered Nanopores

ACS NANO

Lee, Jihang; Du, Chao; Sun, Kai; Kioupakis, Emmanouil; Lu, Wei D.

被引用次数: 147

2016

Very Low-Programming-Current RRAM With Self-Rectifying Characteristics

IEEE ELECTRON DEVICE LETTERS

Zhou, Jiantao; Cai, Fuxi; Wang, Qiwen; Chen, Bing; Gaba, Siddharth; Lu, Wei D.

被引用次数: 103

2016

Data Clustering using Memristor Networks

SCIENTIFIC REPORTS

Choi, Shinhyun; Sheridan, Patrick; Lu, Wei D.

被引用次数: 88

2015

Experimental Demonstration of a Second-Order Memristor and Its Ability to Biorealistically Implement Synaptic Plasticity

NANO LETTERS

Kim, Sungho; Du, Chao; Sheridan, Patrick; Ma, Wen; Choi, ShinHyun; Lu, Wei D.

被引用次数: 490

2015

An Optoelectronic Resistive Switching Memory with Integrated Demodulating and Arithmetic Functions

ADVANCED MATERIALS

Tan, Hongwei; Liu, Gang; Zhu, Xiaojian; Yang, Huali; Chen, Bin; Chen, Xinxin; Shang, Jie; Lu, Wei D.; Wu, Yihong; Li, Run-Wei

被引用次数: 182

2015

Conduction mechanism of a TaOx-based selector and its application in crossbar memory arrays

NANOSCALE

Wang, Ming; Zhou, Jiantao; Yang, Yuchao; Gaba, Siddharth; Liu, Ming; Lu, Wei D.

被引用次数: 45

2015

Utilizing multiple state variables to improve the dynamic range of analog switching in a memristor

APPLIED PHYSICS LETTERS

Jeong, YeonJoo; Kim, Sungho; Lu, Wei D.

被引用次数: 92

2015

Temporal information encoding in dynamic memristive devices

APPLIED PHYSICS LETTERS

Ma, Wen; Chen, Lin; Du, Chao; Lu, Wei D.

被引用次数: 18

2015

Memristive Physically Evolving Networks Enabling the Emulation of Heterosynaptic Plasticity

ADVANCED MATERIALS

Yang, Yuchao; Chen, Bing; Lu, Wei D.

被引用次数: 140

2015

Biorealistic Implementation of Synaptic Functions with Oxide Memristors through Internal Ionic Dynamics

ADVANCED FUNCTIONAL MATERIALS

Du, Chao; Ma, Wen; Chang, Ting; Sheridan, Patrick; Lu, Wei D.

被引用次数: 381

2015

Tuning Resistive Switching Characteristics of Tantalum Oxide Memristors through Si Doping

ACS NANO

Kim, Sungho; Choi, ShinHyun; Lee, Jihang; Lu, Wei D.

被引用次数: 131

2014

Electrochemical dynamics of nanoscale metallic inclusions in dielectrics

NATURE COMMUNICATIONS

Yang, Yuchao; Gao, Peng; Li, Linze; Pan, Xiaoqing; Tappertzhofen, Stefan; Choi, ShinHyun; Waser, Rainer; Valov, Ilia; Lu, Wei D.

被引用次数: 802

2014

Ultralow Sub-1-nA Operating Current Resistive Memory With Intrinsic Non-Linear Characteristics

IEEE ELECTRON DEVICE LETTERS

Gaba, Siddharth; Cai, Fuxi; Zhou, Jiantao; Lu, Wei D.

被引用次数: 40

2014

Retention failure analysis of metal-oxide based resistive memory

APPLIED PHYSICS LETTERS

Choi, Shinhyun; Lee, Jihang; Kim, Sungho; Lu, Wei D.

被引用次数: 50

2014

Crossbar RRAM Arrays: Selector Device Requirements During Write Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Sungho; Zhou, Jiantao; Lu, Wei D.

被引用次数: 132

2014

Single-crystal metallic nanowires and metal/semiconductor nanowire heterostructures

NATURE

Wu, Y; Xiang, J; Yang, C; Lu, W; Lieber, CM

被引用次数: 1098

2004

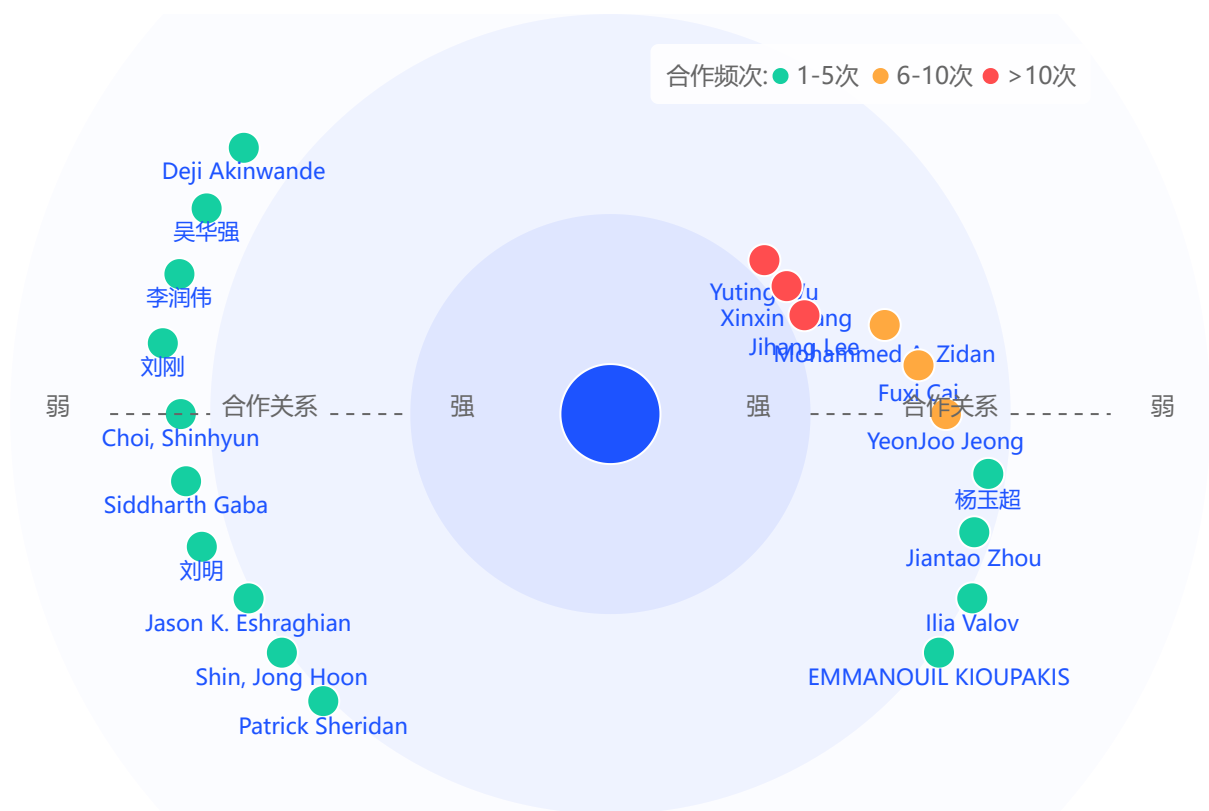
Training Spiking Neural Networks Using Lessons From Deep Learning

Proc.IEEE

Jason K. Eshraghian; Max Ward; Emre O. Neftci; Xinxin Wang; Gregor Lenz; Girish Dwivedi; Mohammed Bennamoun; Doo Seok Jeong; Wei D. Lu

被引用次数: 0

学术合作



Yuting Wu

合作次数: 12

密歇根大学-安娜堡

Xinxin Wang

合作次数: 12

斯坦福大学

Jihang Lee

合作次数: 12

密歇根大学-安娜堡

Mohammed A. Zidan

合作次数: 9

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